Part I:

1) Descriptive Statistics [HW 1.1, 1.2]

- Location: Mean, Median
- Variability: Range, Variance/Std. Deviation

} And how they change under linear transformation.

Book Chapters: 1.1 (only the beginning), 1.3, 1.4 (note - a different def. of variance...)

Things to Know:

- Understanding and computing different descriptive stats.

2) Set Theory [HW 1.3]

- Experiment, Sample Space, Event
- Sets: Empty Set, Subsets, Equality of Sets
- Operations: Union, Intersection, Complement
- DeMorgan's Law
- Properties of two (or more) sets: Disjoint, Mutually Exclusive, Collectively Exhaustive

Book Chapters: 2.1

Things to Know:

- Computing Set Operations
- Proving one set is a subset of some other set.

3) Fundamentals of Probability [HW 1.4, 1.5, 2.1, 2.2, 2.3]

- 3 Axioms of Probability, Properties (e.g. Law of Total Probability)
- Equally Likely Outcomes and Counting Techniques
  + Product Rule for "Sub-Experiments"
  + Permutations and Combinations
- Conditional Probability and Bayes' Rule
- Independence of 2 Events, Pairwise/Mutually Indep. Events

Book Chapters: 2.2, 2.3, 2.4, 2.5

Things to Know:

- Using Axioms of Prob. to compute the prob. of different events
- Counting the # of outcomes in a: i) Sample Space, ii) Event
- Computing Conditional Probabilities, and using Bayes' Rule

4) Discrete Random Variables [HW 2.4, 2.5, 3.1, 3.2, 3.5]

- Def. of a R.V., relation to a sample space
- Characterization: Prob. Mass Fn., Cumulative Dist'n Fn.
  - + Interpretation and Properties
- Expected Value of: X, $g(X)$. Linearity of $E[\cdot]$
- Variance (of a r.v.): Def'n and Properties. Standard Deviation
- Common Families of Discrete RV's: PMF, Parameter(s), Interpretation, Properties
  - + Bernoulli, Geometric, Binomial, Poisson

Book Chapters: 3.1, 3.2, 3.3, 3.4, 3.6 (only the first part)

Things to Know:

- Finding and working with pmf's / cdf's
- Computing $E[\cdot]$ and $Var(\cdot)$ for a discrete r.v. or a function of a discrete r.v.
- How to identify and work with common distributions

5) Continuous Random Variables [HW 3.3, 3.4]

- How they differ from discrete r.v.'s
- Characterization: Prob. Density Fn., Cumulative Density Fn.
  - + Interpretation and Properties
- Expected Value of: X, $g(X)$. Linearity of $E[\cdot]$
- Variance: Def., Properties. Standard Deviation
- Common Families of Cts. R.V.'s: PDF, Parameter(s), Interpretation, Properties

+ Uniform, Exponential, Normal / Gaussian

Book Chapters: 4.1, 4.2, 4.3, 4.4 (first part)

Things to Know:

- Finding and working with pdf's / cdf's
- Computing $E[\cdot]$ and $Var(\cdot)$ for a cts. r.v. or a function of a cts. r.v.
- How to identify and work with common distributions

6) Jointly Distributed Random Variables [HW 4.1 - 4.5]

- Discrete RV's:
  + Joint / Marginal / Conditional PMF's (only if 2)
    - Definitions, Properties
  + Independent RV's (Mutual Indep. if more than two)
- Continuous RV's:
  + Joint / Marginal / Conditional PMF's (only if 2)
  + Independent RV's (Mutual Indep. if more than two)
- Expectation w/ Joint RV's
  + Linearity; $E[X_1 \cdot \ldots \cdot X_n]$ and $Var(X_1 + \ldots + X_n)$ under independence
- Covariance: Def'n, Properties
  + Linearity, $Var(X_1 + X_2)$ without independence
  + Correlation: a normalization of Covariance
- Conditional Expectations: Def'n and "Tower Rule"

Book Chapters: 5.1, 5.2 (first part)

Things to Know:

- Checking if a joint pmf / pdf is valid
- Computing Marginal / Conditional pmf/pdf's
- Checking independence; or building a joint pmf/pdf from indep. r.v.'s
- Computing Expectations - including covariance and conditional exp.'s

Part II:

7) Sampling [HW 5.1 - 5.4]

- iid random variables
- Def: Statistic & Sampling Distribution
- Law of Large Numbers
- Central Limit Thm

Book Chapters: 5.3 - 5.5

Things to Know:

- Using CLT to study the distribution of a sum or average

8) Point Estimation [HW 6.1 - 6.5]

- Framework & Objective
  - "$X_1, \dots X_n$ iid with unknown parameter θ"
  - Def.: Point Estimate/or
- Quality of Estimators: Bias & Variance
- Max. Likelihood Estimation
  - Argmax, Likelihood Fn., ML Estimator

Book Chapters: 6.1, 6.2 (except "Method of Moments")

Things to Know:

- Computing Bias/Variance of an Estimator
- Finding MLE's

9) Confidence Sets [HW 7.1 - 7.3, 7.5, 8.1]

- Motivation vs. Point Estimation – different goals, same setup
- Percentiles & Critical Values
- Def'n of a CI
- Types of CI's: Two-sided, Upper, Lower

- Building CI's with:
  - + Large Sample and Known/Unknown Var.
  - + Small Sample (but Normal) with Known/Unknown Var., t-distribution

Book Chapters: 7.1 - 7.3 (up to pg. 300)

Things to Know:

- Building CI's from probability statements
- Interpretation of CI's

10) Hypothesis Testing [HW 7.4, 8.1 - 8.3, 8.5]

- Motivation - T/F answer, same setup
- Def's: Statistical / Null / Alternative Hypothesis
- Intuition: "Reject H_0 if we see something unlikely under it"
- Type I/II Errors & Tradeoffs
  - + Size/Significance Level, Power of a Test
- Building a Hyp. Test
  - + Test Stat. $T(\dots)$, with known distribution under H_0
  - + Rejection Region R, depending on H_A
  - + Size = Prob. that $T(\dots)$ is in R under H_0
- p-Values & Interpretation
- HT's:
  - + Small Samples from a Normal Pop.
  - + Comparing means of two populations with
    - Large Sample, Known/Unknown Var.
    - Small Sample Normal Pop's, Known/Unknown Var.

Book Chapters: 8.1 - 8.4, 9.1, 9.2

Things to Know:

- How to build your own HT's of a given size and computing power.
- Conducting and Interpreting a Hypothesis Test

11) Regression [HW 8.4]

- Motivation & Context - paired data
- Assumptions - Linear Model, Exogeneity, etc.
- Notion of Fit - MSE
- Finding best fit $\hat{\beta}_0, \hat{\beta}_1$
- Quality of Fit: Residuals, SSE, SST, r^2
- Quality of Estimators: Residual Std. Error, $Var(\hat{\beta}_1)$, Unbiased $\hat{\beta}_1$
- Inference on Slope: CI's and HT's
- Incorrect Assumptions:
  - Assumed "$\mathbb{E}[\varepsilon_i X_i] = 0$" vs. Consequence "$\frac{1}{n}\sum_i \hat{\varepsilon}_i x_i = 0$"
  - Errors Correlated with X's $\rightsquigarrow$ Biased $\hat{\beta}_1$
  - "Correlation vs. Causation"

Book Chapters: 12.1-12.3

Things to Know:

- Deriving & Computing $\hat{\beta}_0$, $\hat{\beta}_1$, r^2, $\hat{\sigma}_\varepsilon^2$, etc.
- Testing $H_0: \beta_1 = 0$

<u>Some Useful Tools:</u>

- Product Rule for Sub-experiments + Counting
- Manipulating expressions like "$\mathbb{P}(c \leq aX + b \leq d)$"
- $\mathbb{E}[aX + bY] = a \cdot \mathbb{E}[X] + b \cdot \mathbb{E}[Y]$ for <u>any</u> X, Y (not necessarily independent)
- $Var(aX+b) = a^2 \cdot Var(X)$
- $Var(aX+bY) = a^2 Var(X) + b^2 Var(Y)$ if X, Y are independent
- $Cov(aX+b, cY+d) = a \cdot c \cdot Cov(X,Y)$
- If $X \sim N(\mu, \sigma^2)$, then $aX+b \sim N(a \cdot \mu + b, a^2\sigma^2)$
- If $X \sim N(\mu_X, \sigma_X^2)$ and $Y \sim N(\mu_Y, \sigma_Y^2)$ are independent, then $aX+bY \sim N(a\mu_X + b\mu_Y, a^2\sigma_X^2 + b^2\sigma_Y^2)$